MAT1190 Lecture Notes

Arky!! :3c

’26 Fall Semester

Contents

1 Day 1: Introduction

This is the first part of a two-semester course. Throughout this class, we will cover Hartshorne Chapter II. To quote Hunter Spink, "if I just went through Hartshorne straight, it would not make any sense; this textbook does not make any sense. As a first exposure, the book is not reasonable. " We will get through scheme theory, though. The goal of this class is to develop scheme theory as a main topic, then show how it applies to a bunch of easier theories. "A lot of people think schemes are so confusing, that you basically have to do 6 to 9 on a completely obsolete theory to show you how it actually matters" (but he won't do that).

Three references that are very useful, outside of Hartshorne.

(i) Ravi Vakil's *The Rising Sea*. This is meant to be a yearlong textbook.

(ii) Jo Harris' *The Geometry of Schemes*. Harris just says "this is schemes, suck it up" then develops a bunch of properties about it.

(iii) Qin Li's *Arithmetic* (something something).

The class will have an *oral* midterm and final (roughly October and December). There will be ungraded exercises (biweekly, as we go?) to turn in throughout the semester; the orals will be based on the easier questions.

This class will start with the theory of manifolds over $\mathbb{R}$, i.e., the classical theory of nice spaces. Consider the following insight on the Gelfand–Kolmogorov theorem: consider $C^0(M, \mathbb{R})$, the ring of continuous functions $M \to \mathbb{R}$ on a manifold M. How do we recover M from $C^0(M, \mathbb{R})$? Colloquially, different points of M see different aspects of any function in $C^0(M, \mathbb{R})$. If we start with a point $p \in M$, we get an $\mathbb{R}$-algebra map $C^0(M, \mathbb{R}) \to \mathbb{R}$ given by evaluating the functions at the given point p; we call this the "evaluation map", denoted

$$\mathrm{ev}_p \colon C^0(M, \mathbb{R}) \to \mathbb{R}, \quad f \mapsto f(p).$$

Specifically, $\mathrm{ev}_p \in \mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M, \mathbb{R}), \mathbb{R})$. What does it mean for ev_p to be an $\mathbb{R}$-algebra map? As an example, while $\mathrm{Hom}(\mathbb{C}, \mathbb{C}) = \mathrm{Aut}(\mathbb{C})$ contains "uncountably many horrible things", we see that $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(\mathbb{C}, \mathbb{C})$ consists of homomorphisms that fix $\mathbb{R}$, i.e., it only admits the identity and complex conjugation.

Claim 1.1. $\{\mathrm{ev}_p\}_{p \in M}$ are the only elements of $\mathrm{Hom}_{\mathbb{R}-\mathrm{alg}}(C^0(M, \mathbb{R}), \mathbb{R})$.

We will do the proof for $M = \mathbb{R}$, then see that the proof follows analogously for several other examples of M as well.

Proof. Assume we have an $\mathbb{R}$-algebra map $\Phi \colon C^0(M, \mathbb{R}) \to \mathbb{R}$ that sends the identity function x to a, i.e., $x \mapsto a$. We claim that this map is ev_a; if not, then there is some function $g \in C^0(M, \mathbb{R})$ such that $g \not\mapsto g(a)$ under Φ. Indeed, consider $\ker(\Phi)$; we see that $x - a \in \mathrm{Ker}(\Phi)$; now, consider the function

$$(x - a)^2 + (g(x) - b)^2,$$

where $b \coloneqq \Phi(g(x))$. This function was constructed to never vanish; the only way for it to be zero would be for $x = a$ and $g(x) = \Phi(g(x))$, but this never happens, so it is nonzero everywhere. However, $x - a \in \ker \Phi$ and $g(x) - b \in \ker \Phi$, so

$$(x - a)^2 + (g(x) - b)^2 \in \ker \phi,$$

whence it is a unit; this would mean

$$1 = \frac{1}{(x - a)^2 + (g(x) - b)^2} \cdot \left((x - a)^2 + (g(x) - b)^2\right) \in \ker \Phi,$$

which is contradictory. □

Following Grothendieck, let R be any ring, and consider Spec R, which is a set of points with a topology, and is a locally ringed space (i.e., local functions "glue" to form global functions). We will rigorously define Spec R later. The key insight about manifolds is that, since they consist of charts that are compatible to each other locally, it allows us to "glue" data from functions together. Another important thing to know about locally ringed spaces is that if a function is non-vanishing everywhere, then it is invertible. Together, this lets you recover a topology on your ring R. Note that this stuff is still intentionally hand-wavey for now.[1]

Consider the polynomials over the real numbers, $\mathbb{R}[x]$. It is a ring. What would happen if we tried to reconstruct the space according to the procedure described earlier? Let's look at the $\mathbb{R}$-algebra homomorphisms from $\mathbb{R}[x] \to \mathbb{R}$; then we get that it is $\mathbb{R}$, since $x \mapsto a \in \mathbb{R}$, and there is probably some way to proceed from here.

There is, however, a reason we don't do algebraic geometry over the real numbers; that reason being, the process would suggest that $\mathbb{R}[x]$ is that ring of functions on the real line. If we looked at $x^2 + 1 \in \mathbb{R}[x]$, we would see that it's a polynomial that doesn't vanish anywhere and not invertible. This makes these "space-recovery" arguments impossible, specifically the part where a "non-vanishing function is invertible". There are two solutions to this behavior... either you invert functions which don't vanish everywhere, so you map $\mathbb{R} \to \mathbb{R} \setminus \{0\}$, or you can add points for every possible evaluation to every field. For example, consider

$$\mathrm{ev}_i \colon \mathbb{R}[x] \to \mathbb{C}, \quad x \mapsto i;$$

, we would see that $x^2 + 1 \mapsto 0$. Next time, we will define that $\mathrm{Spec}(R)$ is built out of all homomorphisms from $\mathbb{R}$ to fields, modded out by "some stuff".

[1] "you... uncover something so unbelievably terrible that i can't even begin to describe it. something that's going to ruin your life for the next... twelve weeks!"